MeshAlive

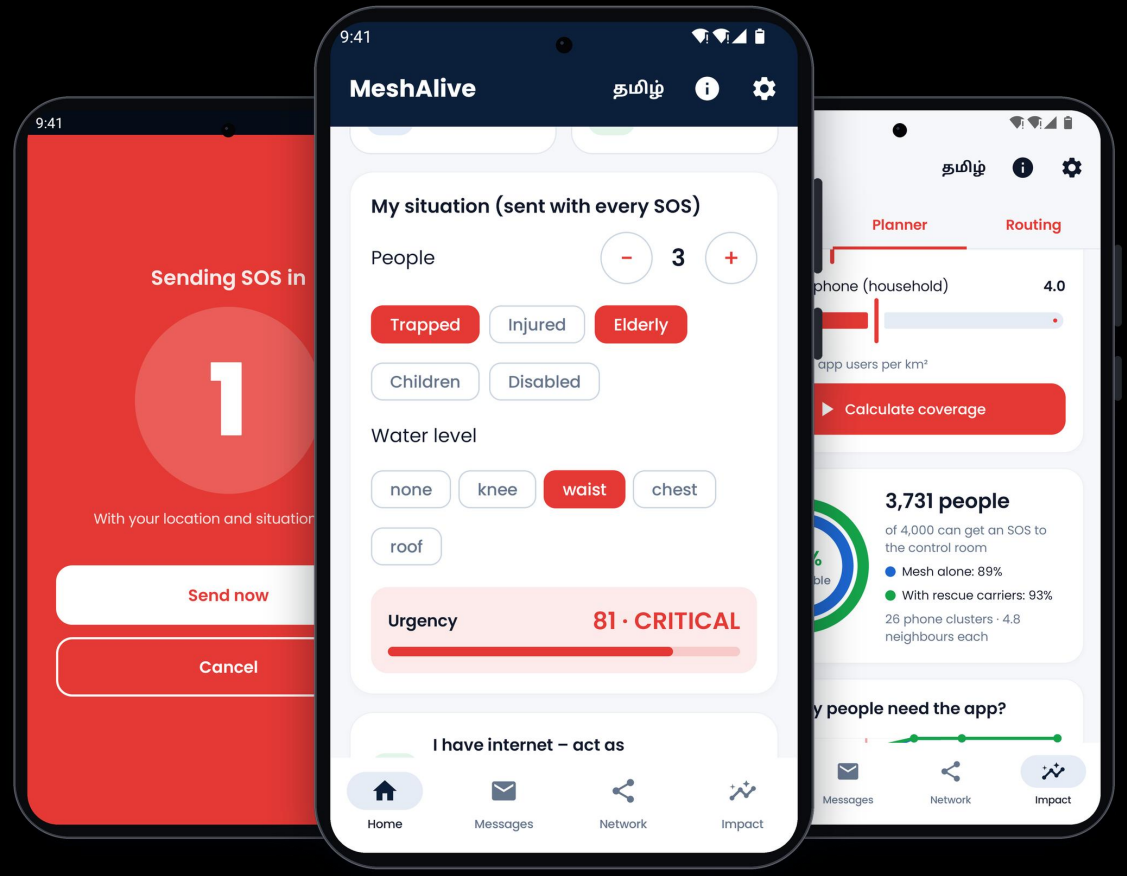
Help reaches you, even when the network doesn’t.

An offline disaster-response network. When cyclones and floods take down mobile towers, SOS messages travel phone to phone and through rescue boats to the district control room, and help comes back the same way.

Theme 05 · Surakshit Bharat

Track A · Product

Working prototype



THE PROBLEM

When the towers go down, rescue goes blind

Every cyclone and flood season, districts lose mobile coverage for hours or days, as in the 2015 Chennai floods and Cyclone Michaung in 2023. That is exactly when people most need help.

People can’t call for help

Trapped families, the elderly and patients can’t call, text or share their location.

Control rooms work blind

No view of who needs help, where they are, how many people, or how urgent.

Chat apps aren’t enough

They don’t reach rescuers or prioritise, and they drain the phones that relay for others.

THE SOLUTION

A complete rescue loop on phones people already own

1

Ask for help

One-tap SOS with GPS and a situation report, scored 0–100 for urgency.

2

Phones relay

Bluetooth and Wi-Fi mesh. Battery-aware routing; most urgent first.

3

Rescuers carry

Boats and NDRF teams carry stuck SOS out of dead zones.

4

Control room sees

The first phone with signal uploads to a live, urgency-sorted map.

5

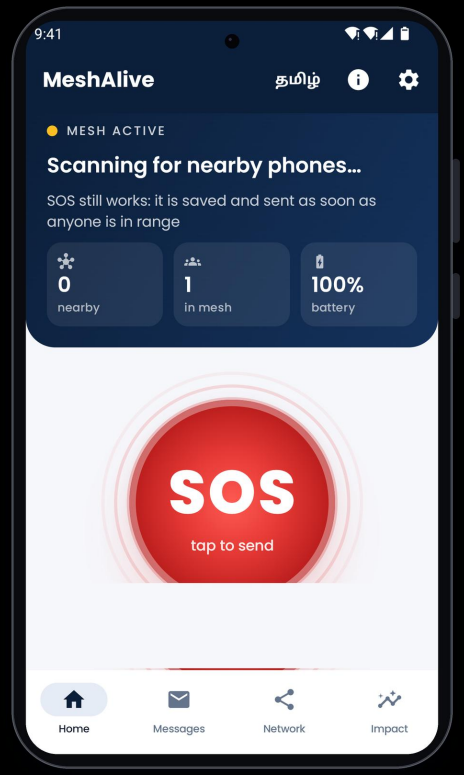
Help comes back

Dispatch and evacuation alerts reach phones with no signal.

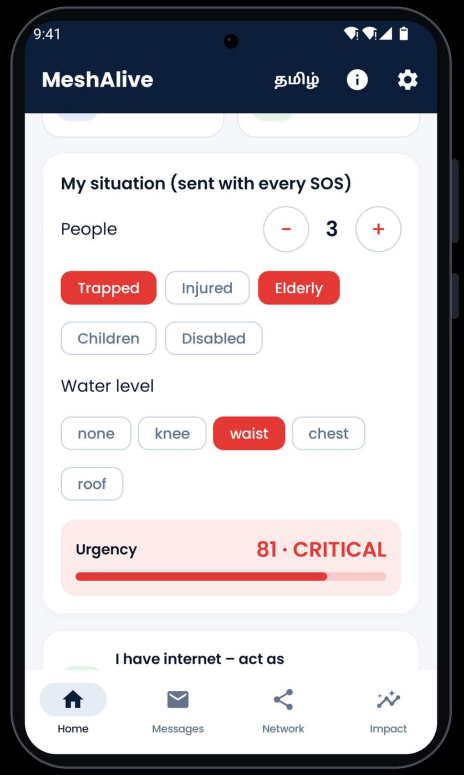
No towers · No internet · No extra hardware · Tamil and English

THE APP

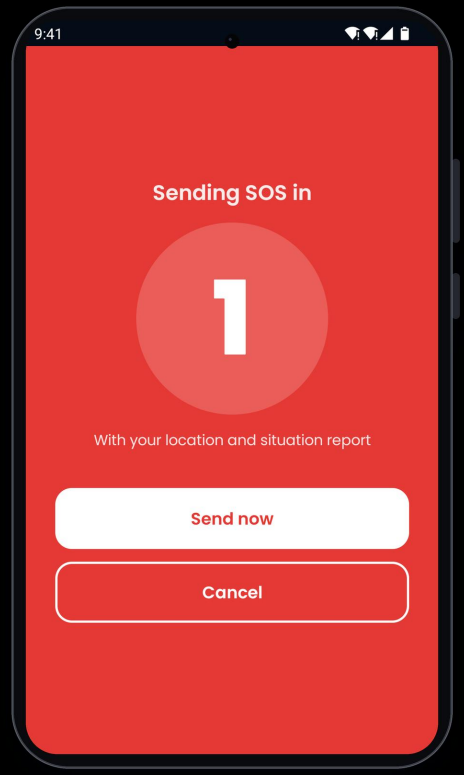
Built for the worst day of someone’s life



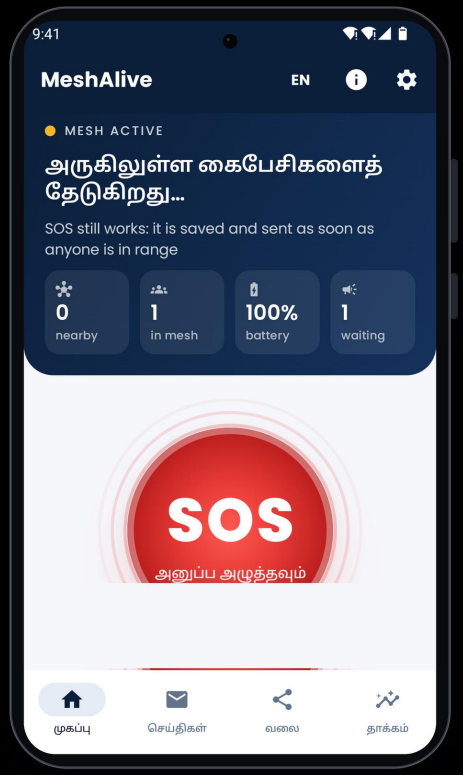
Live mesh status and one-tap SOS



Situation report → urgency 81 · Critical



3-second countdown: no false alarms



Fully in Tamil

DISTRICT CONTROL ROOM

Every SOS on one live map, and help sent back

- **Sorted by urgency**

  Critical cases first, with people count and route.

- **Dispatch, and the victim knows**

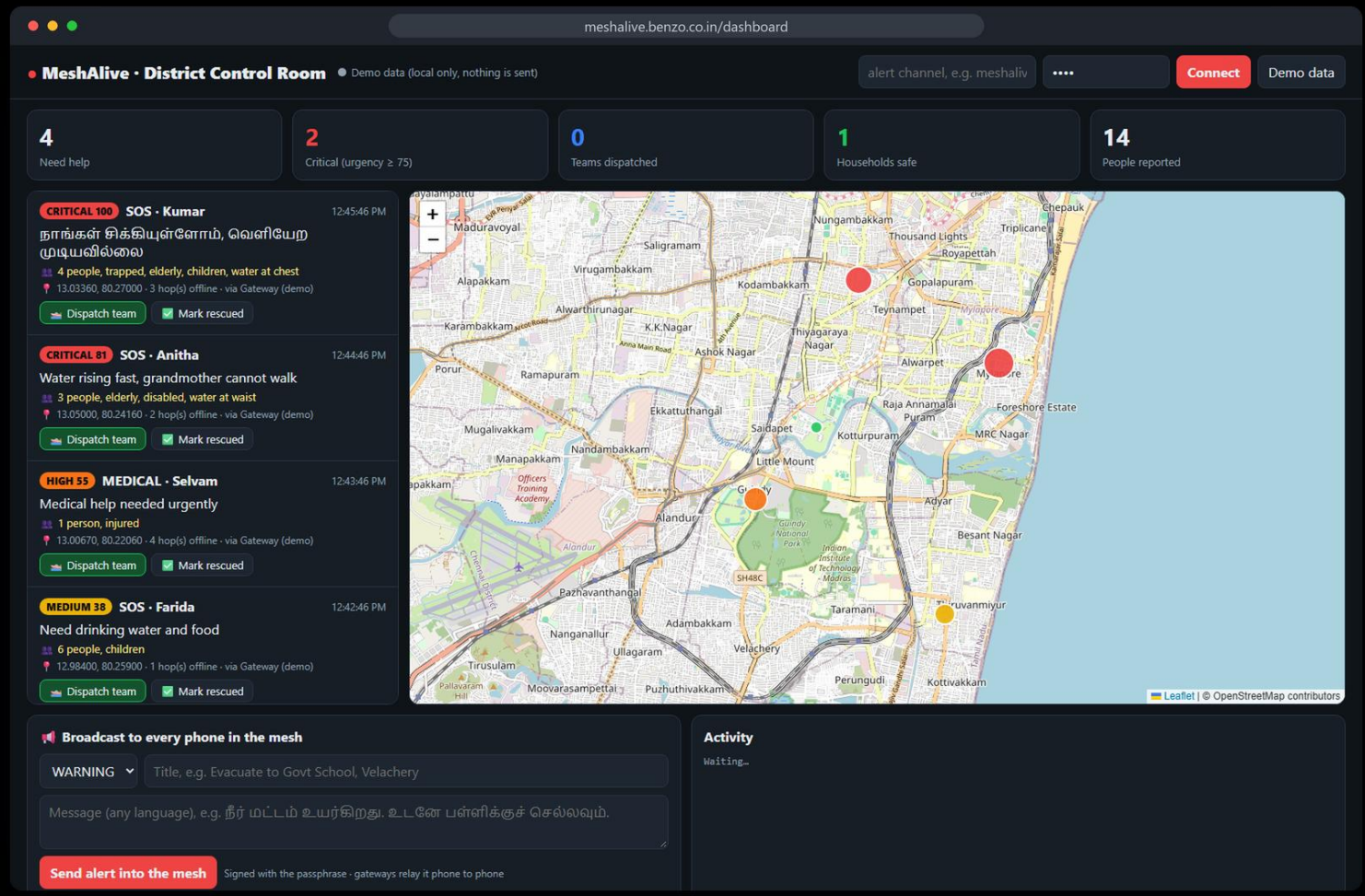
  “Rescue team on the way” reaches phones with no signal.

- **Evacuation broadcasts**

  In any language, to every phone in the mesh.

- **Signed commands**

  Phones reject anything not signed by the control room.



WHY IT'S DIFFERENT

A rescue system, not a chat app

	Chat apps (Bridgefy, Briar)	LoRa radios (Meshtastic, goTenna)	MeshAlive
Reaches rescuers	No	No	**Live district dashboard**
Help back to the victim	No	No	**Dispatch + evacuation alerts**
Prioritises urgent cases	No	No	**Triage score 0–100**
No phone in range	Message waits	Message waits	**Rescue carriers carry it out**
Battery	Flooding drains phones	Flooding	**Battery-aware routing**
Extra hardware	None	₹2,500+ per person	**None**
Language	English	English	**Tamil and English**

UNDER THE HOOD

How MeshAlive works

Discovery	Phones find and connect to each other over Bluetooth and Wi-Fi (Google Nearby Connections).
Network map	Each phone shares battery, neighbours and internet status; every phone rebuilds the mesh map.
Routing	Battery-aware Dijkstra: link quality + battery penalty + hop cost, one run via a virtual super-node.
Reliability	Urgency priority queue, store-and-forward, duplicate filter, hop limit and delivery receipts.
Carriers & gateways	Custody transfer to rescue carriers; any phone with signal uploads what it carries.
Security	AES-GCM encrypted messages; HMAC-SHA256 signed control-room commands.

EVIDENCE

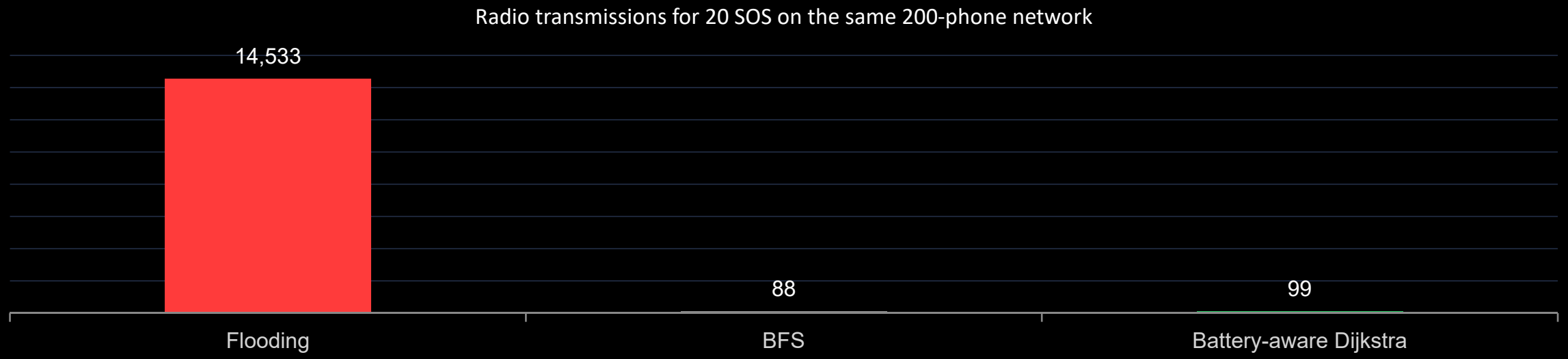
Built, tested and measured

100%
SOS delivered (200-phone simulation)

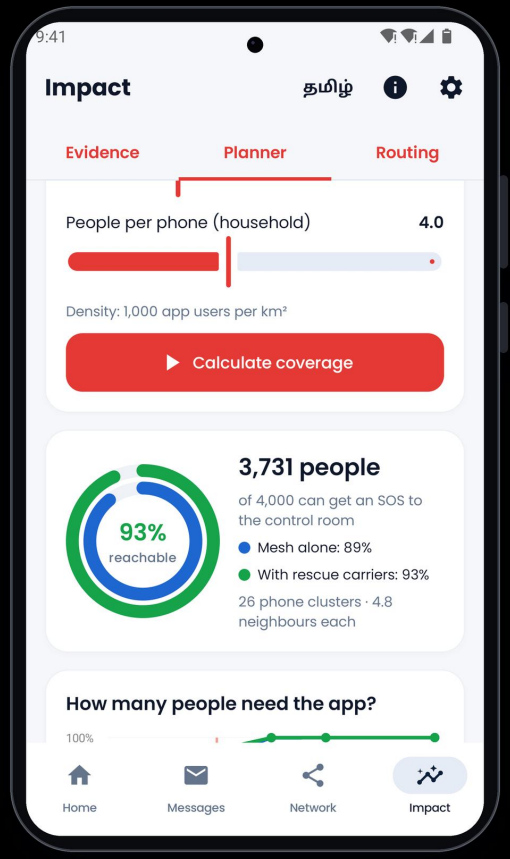
147×
fewer radio transmissions than flooding

2×
battery left on the weakest phone

49
automated tests



Who benefits, what it costs, how it grows



District planner (the app's own routing engine)

1 km² ward, 1,000 app users, 4 people per phone: about 3,700 of 4,000 people can get an SOS to the control room (model, to be validated in the pilot).

₹0 — **Citizens**

Free app on their own phone

₹0 — **Rescue carriers**

Phones crews already carry

₹0 — **Control room**

Any browser on an existing PC

DDMA — **Rollout**

Pre-monsoon install & awareness drives

PATH TO DEPLOYMENT

From one ward to every district

PHASE 1 · 3 MONTHS

Field pilot

One flood-prone ward with the DDMA, NDRF/SDRF boats as carriers, dashboard in the district EOC. Measure delivery rate and SOS-to-dispatch time.

PHASE 2

District rollout

Pre-monsoon install drives via schools, colleges, panchayats, ULBs and SHGs, guided by the planner’s adoption targets.

PHASE 3

State & national

Distribution through the SDMA; state EOC integration; alerts aligned with the national Common Alerting Protocol; more Indian languages.

WHAT WE NEED

Help us take MeshAlive to the field

- Pilot partnership with a district administration and its DDMA
- NDRF/SDRF and boat operators to field-test carrier mode
- Mentorship in disaster operations and public-sector deployment
- Guidance on security review and Common Alerting Protocol integration
- Pilot funding: alert server, test devices, awareness drives
- IP guidance and incubation support

MeshAlive: because when the network fails, help shouldn’t.

meshalive.benzo.co.in · github.com/yogeshwardev/MESHALIVE-